APT & DPKG INVESTIGATION REPORT

Chapter 9 — Lab 2: Package Troubleshooting & Security Investigation

ENVIRONMENT	TARGET PACKAGE	DIFFICULTY	EXECUTION MODE
Kali Linux (kali-rolling)	**curl (8.20.0-5)**	**Beginner–Intermediate**	**Safe / Simulated**

1. OVERVIEW & INVESTIGATION SCOPE

This report documents a comprehensive Linux system administration and cybersecurity investigation focusing on Debian package management using dpkg and Advanced Package Tool (apt). All operations were executed on Kali Linux. Simulations and non-destructive status checks were utilized to analyze package health, candidate versions, repository updates, upgrade impact, and purge behaviors without introducing unwanted state changes to the target system.

2. TASK 1 & TASK 2 — PACKAGE AUDIT & INSPECTION

Task 1 — Package Database Audit

```
$ sudo dpkg --audit    READ-ONLY
```

Result: Returned clean execution (no output). This confirms there are no broken half-installed, half-configured, or unmerged packages pending repair in the local dpkg database.

Task 2 — Package Verification & Path Location

```
$ dpkg -s curl    READ-ONLY
```

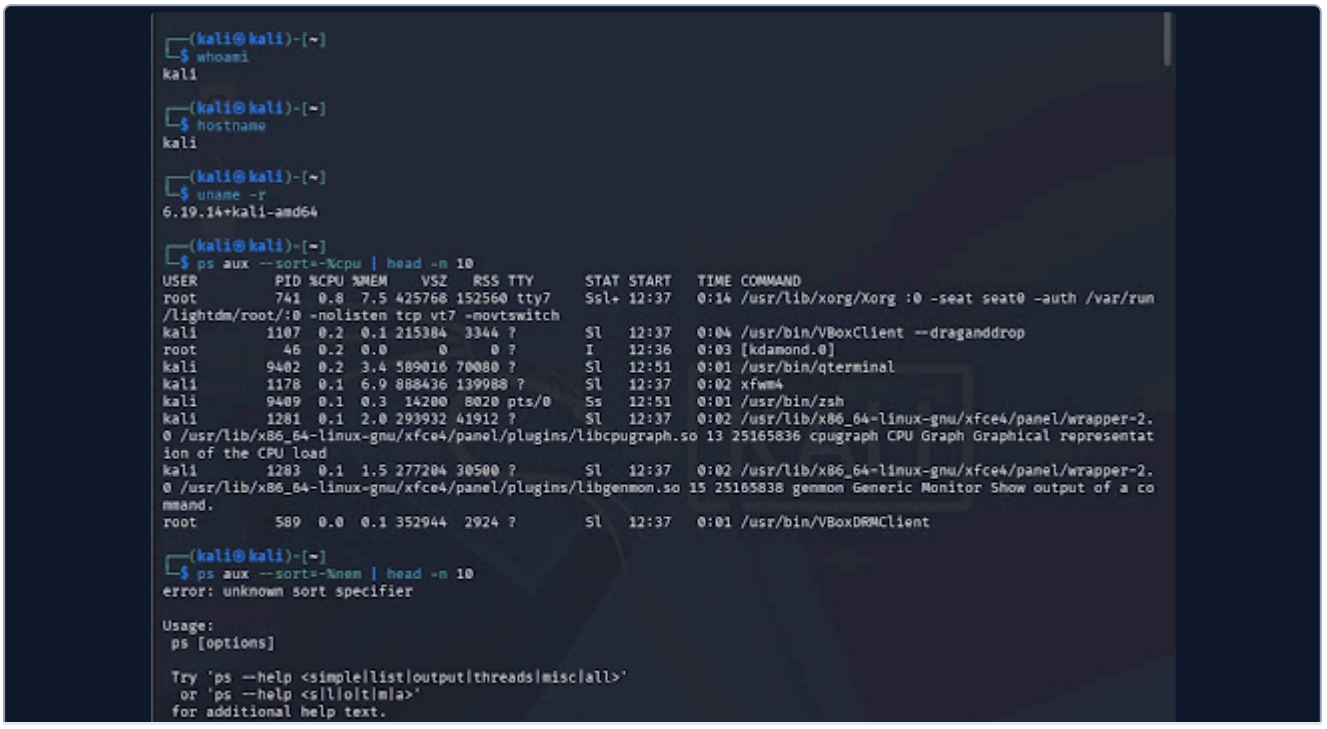
Findings: `Status: install ok installed` | Installed Version: `8.20.0-5` | Architecture: amd64.

```
$ command -v curl    READ-ONLY
```

Binary Location: `/usr/bin/curl` — Confirms executable exists in standard user PATH.

```
$ apt-cache policy curl    READ-ONLY
```

Policy Summary: Installed Version: `8.20.0-5` (Priority 100 in `/var/lib/dpkg/status`)
Candidate Version: `8.22.0-1` (Priority 500 from `http://http.kali.org/kali kali-rolling/main`)
Mission Conclusion: `curl` is installed. The candidate version available for upgrade is `8.22.0-1`.



3. TASK 3 & TASK 4 — REPOSITORY HEALTH & UPGRADE SIMULATIONS

Task 3 — Repository Health Check

```
$ sudo apt update    READ-ONLY REFRESH
```

Observation: Fetched 77.4 MB in 14s (5,375 kB/s) across Kali rolling repositories (`main`, `contrib`, `non-free`, `non-free-firmware`). No signature errors, expired keys, or DNS network failures were observed. Total upgradable packages identified: **1,530 packages**.

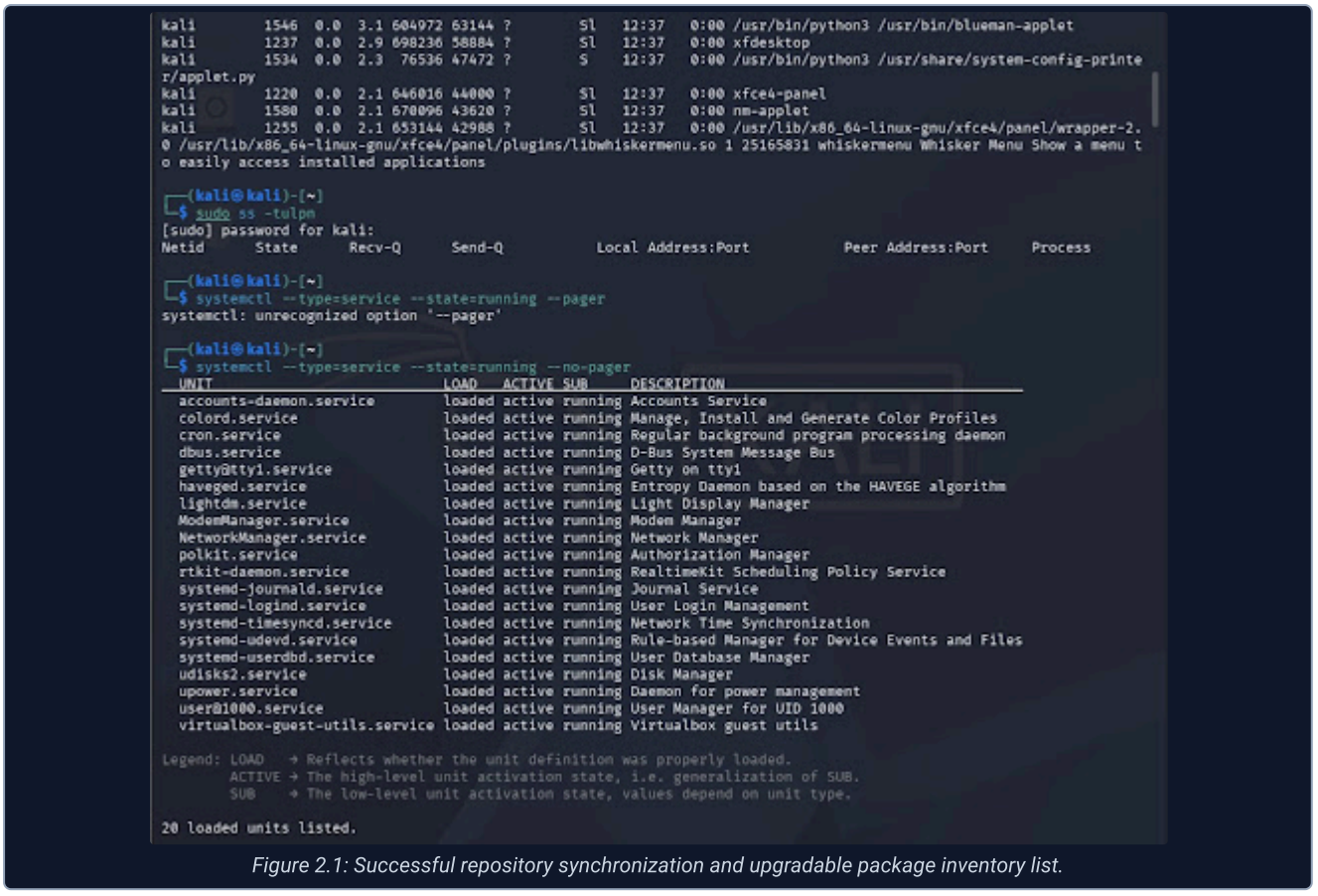


Figure 2.1: Successful repository synchronization and upgradable package inventory list.

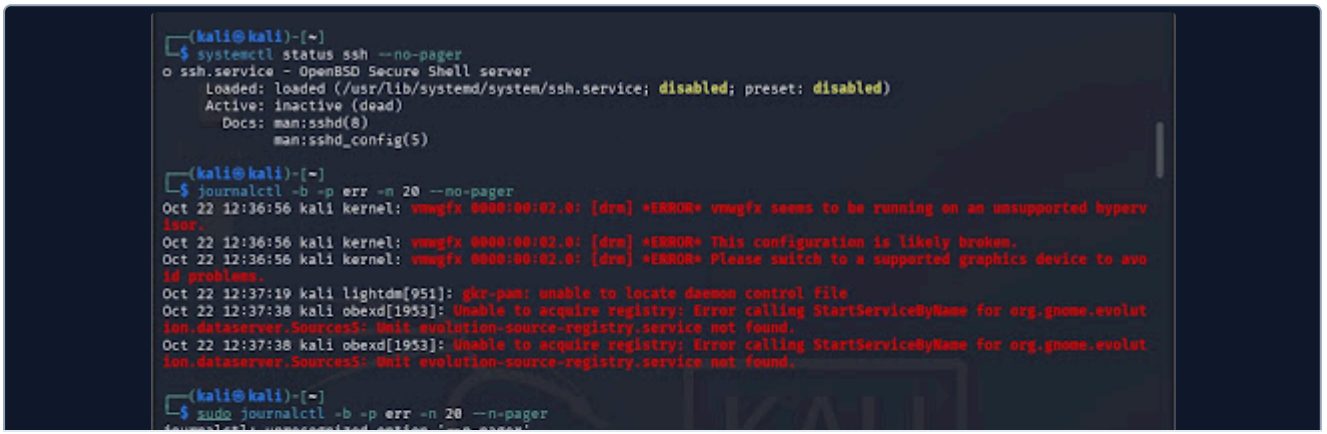
Task 4 — Upgrade Simulation & Comparison

```
$ apt list --upgradable    READ-ONLY
```

Lists pending updates including core system binaries (`bash`, `apt`, `binutils`, `chromium`, `curl`).

```
$ apt upgrade --simulate    SIMULATION
```

Simulation Summary: Upgrading: 1,492 | Installing: 88 | Removing: 0 | Not Upgrading: 38.
Standard upgrade avoids removing installed packages or resolving breaking dependency shifts.



4. TASK 5 — REMOVAL VS. PURGE COMPARATIVE ANALYSIS

```
$ apt remove --simulate curl    SIMULATION
```

Analysis: Simulates removing `curl` binary along with 16 dependent meta-packages/tools (e.g., `kali-linux-default`, `metasploit-framework`, `wpscan`, `legion`). Configuration files are retained.

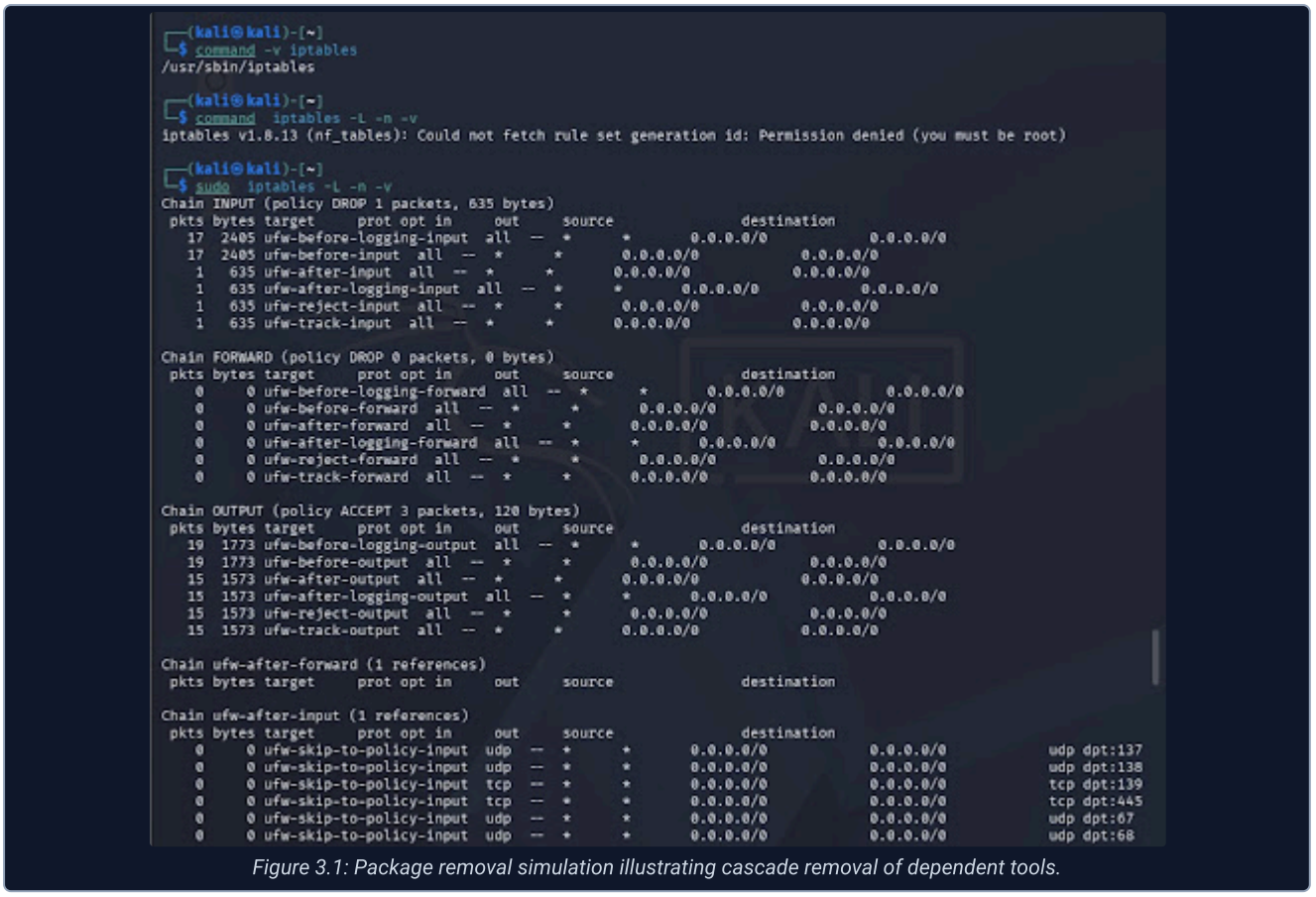
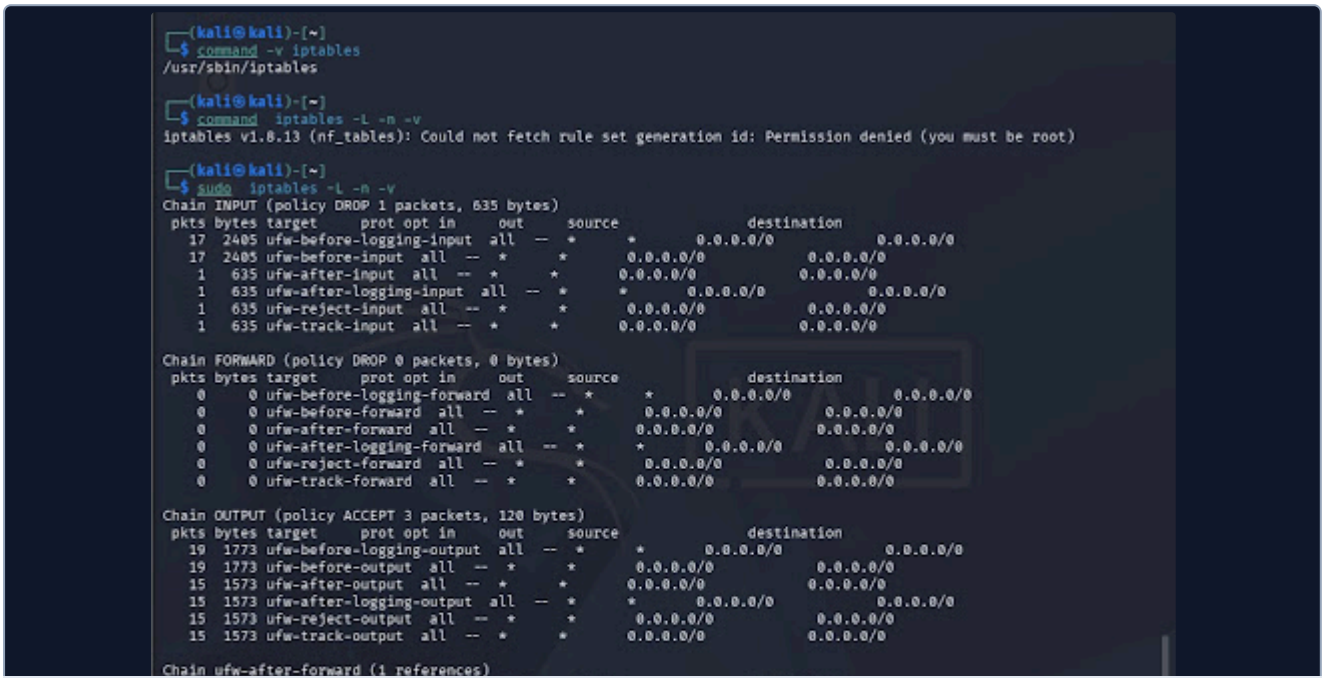


Figure 3.1: Package removal simulation illustrating cascade removal of dependent tools.

```
$ apt purge --simulate curl    SIMULATION
```

Analysis: Indicated by `Purg` flag in execution logs (e.g., `Purg curl [8.20.0-5]`). Instructs APT to delete the package AND erase all associated package-managed system configuration files.



6. FINAL CHALLENGE ASSESSMENT QUESTIONS & ANSWERS

Question 1: What does 'sudo apt update' do?

✔ **Answer: Refreshes repository package lists**

Explanation: Downloads the latest package index files from configured mirrors. It does not install or upgrade packages on disk.

Question 2: Which command checks the installed status of a package?

✔ **Answer: dpkg -s curl**

Explanation: Queries the internal dpkg database for installation state, version, architecture, and maintainer info.

Question 3: Why use 'apt full-upgrade --simulate'?

✔ **Answer: To inspect proposed changes without performing the upgrade**

Explanation: Perform dry-run evaluation to preview package removals or major dependency upgrades without making live system changes.

Question 4: What is the key difference between remove and purge?

✔ **Answer: Purge also removes package-managed configuration files where applicable**

Explanation: While 'remove' leaves configuration files intact in /etc for future re-installations, 'purge' cleans them entirely.

7. KEY TECHNICAL SUMMARY & BEST PRACTICES

Utility	Primary Use Case	Safety Level
`dpkg --audit`	Detect incomplete package states & broken installations.	SAFE / READ-ONLY
`apt-cache policy`	View installed vs available candidate versions & priorities.	SAFE / READ-ONLY
`apt update`	Refresh metadata cache from remote Debian/Kali repositories.	SAFE / REFRESH
`--simulate` flag	Test complex dependency shifts, removals, and upgrades.	100% DRY-RUN